

Online Appendix to “Optimal Disclosure Windows”

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OA.1 Benchmark: Perfectly Informed Agent

As a benchmark, consider the scenario in which the agent is informed of whether the state is $\theta = 0$ or $\theta = 1$ at time 0. As disclosure starts later, the stopping game converges to the benchmark case where the agent is perfectly informed. As mentioned in [Proposition 3](#), the waiting time in the continuation stopping game is bounded above by $\bar{w}^*$, which is the waiting time in this benchmark. [Proposition OA.1](#) formalizes this argument.

Proposition OA.1. *In the continuation game, there exists $0 < \bar{w}^* < \infty$ such that the informed agent randomizes over stopping times in $[0, \bar{w}^*]$ and uninformed agent stops with probability 1 at $\bar{w}^*$.*

The proof is the same as the proof of [Theorem 1.B](#) (ii) as is thus omitted.

OA.2 Omitted Proofs for [Section 3](#)

OA.2.1 Proof of Lemma 19

Take the derivative of $V^*(t)$ with respect to t ,

$$V^{*'}(t) = \frac{e^{\lambda t - r w^*(t)} \lambda \mu (1 - \mu) - e^{\lambda t - r w^*(t)} \mu (r(1 - \mu) + e^{\lambda(t + w^*(t))} (r + \lambda) \mu) w^{*'}(t)}{(1 + (e^{\lambda(t + w^*(t))} - 1) \mu)^2}.$$

It suffices to show

$$e^{\lambda t - r w^*(t)} \lambda \mu (1 - \mu) - e^{\lambda t - r w^*(t)} \mu (r(1 - \mu) + e^{\lambda(t + w^*(t))} (r + \lambda) \mu) w^{*'}(t) < 0,$$

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which is equivalent to showing

$$w^{*'}(t) > \frac{\lambda(1-\mu)}{r(1-\mu) + e^{\lambda(t+w^*(t))}(r+\lambda)\mu}.$$

Recall that

$$w^{*'}(t) = \frac{\beta e^{(\beta+r)w^*(t)} + r}{\beta(e^{(\beta+r)w^*(t)} - 1)} \frac{\lambda(1-\mu)}{r(1-\mu) + e^{\lambda(t+w^*(t))}(r+\lambda)\mu}.$$

The first term is greater than 1 and the second term is positive. The result follows.

OA.2.2 Proof of Lemma 20

I prove the case for the uninformed agent, the informed agent follows from the same algebra.

By definition, $Y(t|t_0)$ is given by

$$\begin{aligned} Y(t|t_0) = & (1 - \rho(t_0)) \int_{t_0}^t \lambda e^{-\lambda(s-t_0)} e^{-r(s-t_0)} V^*(s) ds \\ & + ((1 - \rho(t_0))e^{-\lambda(t-t_0)} + \rho(t_0)) e^{-r(t-t_0)} U^*(t). \end{aligned}$$

Take the derivative with respect to t ,

$$\begin{aligned} Y'(t|t_0) = & e^{-r(t-t_0)} \frac{e^{r(t-t_0)} \rho(t)}{(1 - \rho(t_0))e^{-\lambda(t-t_0)} + \rho(t_0)} \\ & \cdot (\rho'(t) (V^*(t) - U^*(t)) + \rho(t) (U^{*'}(t) - rU^*(t))). \end{aligned}$$

where the first line is strictly positive and the second line is proportional to $y(t)$. The result follows.

OA.2.3 Proof of Lemma 21

Take the derivative of $Y(t)$ with respect to t . After simplifying, the derivative is proportional to

$$\lambda(1-\mu)e^{-\lambda t} (V^*(t) - U^*(t)) + ((1-\mu)e^{-\lambda t} + \mu) (U^{*'}(t) - rU^*(t)).$$

Recall that the derivative of the best-case scenario payoff $Y_0(t)$ is $\rho'(t) - r\rho(t)$, which, by assumption, is negative. Because $V^*(t) - U^*(t) < 0$, that is, the informed agent's equilibrium continuation payoff is lower than the uninformed, it suffices to show if $\rho'(t) - r\rho(t) < 0$ for all t , $U^{*'}(t) - rU^*(t) < 0$ for all t .

The rest of the proof proceeds by brute force calculations. One can explicitly write down the expression for $U^{*'}(t) - rU^*(t)$. After simplifying,

$$\begin{aligned}
U^{*'}(t) - rU^*(t) = & \frac{e^{-(\beta+r)w^*(t)}}{(\beta - \lambda)(\rho(t)(e^{\lambda w^*(t)} - 1) + 1)} r\rho(t) (\lambda\rho(t)e^{\lambda w^*(t)} - e^{\beta w^*(t)}(\beta - \lambda + \lambda\rho(t))) \\
& + \rho'(t) \frac{e^{-(\beta+r)w^*(t)}}{(\beta - \lambda)(\rho(t)(e^{\lambda w^*(t)} - 1) + 1)^2} \\
& \cdot (\lambda\rho(t)(e^{\beta w^*(t)} - e^{\lambda w^*(t)})(1 + \rho(t)e^{\lambda w^*(t)} + 1 - \rho(t)) + (\beta - \lambda)e^{\beta w^*(t)}) \\
& + w^{*'}(t) \frac{e^{-rw^*(t)}\rho(t)}{(\beta - \lambda)(\rho(t)(e^{\lambda w^*(t)} - 1) + 1)^2} \\
& \cdot \left(r(\lambda - \beta) + \rho(t)(r(\beta - 2\lambda + \lambda\rho(t))) + \rho(t)\lambda e^{(\lambda-\beta)w^*(t)}(\beta - \lambda + r) \right. \\
& + \lambda e^{(\lambda-\beta)w^*(t)}\rho(t)^2(\lambda - \beta - r - (\lambda + r)e^{\beta w^*(t)} + re^{\lambda w^*(t)} + \beta e^{\lambda w^*(t)}) \\
& \left. + e^{(\lambda-\beta)w^*(t)}\rho(t)(\lambda - \beta)(\lambda + r)e^{\beta w^*(t)} \right)
\end{aligned}$$

The expression for $w^{*'}(t)$ is given by (23). It can be verified that for each t , whenever $\rho'(t) - r\rho(t) < 0$, the right-hand side is negative for all w^* . So with the equilibrium waiting time $w^*(t)$, $U^{*'}(t) - rU^*(t) < 0$.

OA.3 Proofs for Section 5

OA.3.1 Proof of Proposition 4

Let $q_D(w)$ denote the decision maker's belief that $\theta = 1$ conditional on no signals arriving in $[0, w]$. By Bayes' rule,

$$q_D(w) = \frac{\eta}{\eta + e^{-\lambda w}(1 - \eta)}. \quad (\text{OA.1})$$

Given this belief $q_D(w)$, from the perspective of the beginning of the continuation stopping game, denote the decision maker's expected payoff from stopping at w by $D(w)$, then

$$\begin{aligned}
D(w) = & (1 - \eta) \int_0^w e^{-rs} \lambda e^{-\lambda s} e^{-\beta s} ds \\
& + (1 - \eta) \int_0^w e^{-rs} e^{-\lambda s} \beta e^{-\beta s} (1 - (q_D(s) - 0)^2) ds \\
& + (1 - \eta) e^{-rw} e^{-\lambda w} e^{-\beta w} (1 - (q_D(w) - 0)^2) \\
& + \eta \int_0^w e^{-rs} \beta e^{-\beta s} (1 - (q_D(s) - 1)^2) ds \\
& + \eta e^{-rw} e^{-\beta w} ((1 - (q_D(w) - 1)^2)).
\end{aligned} \quad (\text{OA.2})$$

Take the derivative of $D(w)$ with respect to w . After simplifying,

$$D'(w) \propto \lambda(1 - q_D(w))q_D(w)^2 - r(1 - (1 - q_D(w))q_D(w)). \quad (\text{OA.3})$$

Define

$$R(w; \eta) := \left(\frac{e^{-\lambda w} (1 - \eta) + \eta}{e^{-\lambda w} (1 - \eta)} + \left(\frac{\eta + e^{-\lambda w} (1 - \eta)}{\eta} \right)^2 \right)^{-1}. \quad (\text{OA.4})$$

Rearrange the right-hand side of (OA.3) and substitute (OA.1) for $q_D(w)$, $D'(w) \leq 0$ if and only if $R(w; \eta) \leq r/\lambda$.

Claim OA.1. For any w and η , $\partial R(w; \eta)/\partial w$ and $\partial R(w; \eta)/\partial \eta$ have the same sign.

Proof. To facilitate the proof, note that $q_D(w)$ depends on both w and η . With a slight abuse of notation, denote (OA.1) by $q_D(w; \eta)$. Then $R(w; \eta)$, defined in (OA.5), can be written as

$$R(w; \eta) = \left(\frac{1}{1 - q_D(w; \eta)} + \frac{1}{q_D(w; \eta)^2} \right)^{-1}. \quad (\text{OA.5})$$

Then

$$\frac{\partial R(w; \eta)}{\partial w} = \frac{\partial R(w; \eta)}{\partial q_D} \frac{\partial q_D(w; \eta)}{\partial w} \quad \text{and} \quad \frac{\partial R(w; \eta)}{\partial \eta} = \frac{\partial R(w; \eta)}{\partial q_D} \frac{\partial q_D(w; \eta)}{\partial \eta}.$$

By (OA.1), $\partial q_D(w; \eta)/\partial w > 0$ and $\partial q_D(w; \eta)/\partial \eta > 0$. The result follows. $\square$

Claim OA.2. $R(w; \eta)$ has the following properties.

- (i) For all $0 < \eta < 1$, $R(w; \eta)$ is single-peaked in w : there exists a unique $w_R^*(\eta) = \arg \max_w R(w; \eta)$ with $w_R^*(\eta) \geq 0$ such that $R(w; \eta)$ is strictly increasing in w for $w \leq w_R^*(\eta)$ and strictly decreasing in w for $w > w_R^*(\eta)$.
- (ii) There exists a unique $0 < \eta_R < 1$ such that
 - (a) for all $\eta < \eta_R$, $w_R^*(\eta) > 0$, $w_R^{*\prime}(\eta) < 0$, and $R(w_R^*(\eta); \eta)$ is constant in η ;
 - (b) for all $\eta \geq \eta_R$, $w_R^*(\eta) = 0$ and $R(w_R^*(\eta); \eta) = R(0; \eta)$ is decreasing in η ;
 - (c) Moreover, $w_R^*(\eta)$ is continuous for all $0 < \eta < 1$. That is, $\lim_{\eta \rightarrow \eta_R} w_R^*(\eta) \rightarrow 0$.
- (iii) For all $0 < \eta < 1$, $\lim_{w \rightarrow \infty} R(w; \eta) \rightarrow 0$.

Proof. These properties follow directly from operating on (OA.4).

Specifically, for (ii) (a) and (c), if $w_R^*(\eta)$ is interior, namely, $w_R^*(\eta) > 0$, $w_R^*(\eta)$ has a closed form which can be obtained by solving $\partial R(w; \eta)/\partial w = 0$. It follows from the functional form of $w_R^*(\eta)$ that $w_R^{*\prime}(\eta) < 0$ and $\lim_{\eta \rightarrow \eta_R} w_R^*(\eta) \rightarrow 0$. By definition,

$$\frac{dR(w_R^*(\eta); \eta)}{d\eta} = \frac{\partial R(w_R^*(\eta); \eta)}{\partial \eta} + \frac{\partial R(w_R^*(\eta); \eta)}{\partial w} \frac{dw_R^*(\eta)}{d\eta}.$$

The second term on the right-hand side is zero because $\partial R(w_R^*(\eta); \eta)/\partial w = 0$. By [Claim OA.1](#), the first term $\partial R(w_R^*(\eta); \eta)/\partial \eta = 0$. Therefore, $dR(w_R^*(\eta); \eta)/d\eta = 0$, which means $R(w_R^*(\eta); \eta)$ is constant in η .

For (ii) (c), it follows from the definition ([OA.4](#)) that $R(0; \eta) = \left(\frac{1}{1-\eta} + \frac{1}{\eta^2}\right)^{-1}$ and is decreasing in η . $\square$

Part (i). It follows from [Claim OA.2](#) (ii) that $R(0; \eta_R) = \max_{\eta \in (0,1)} R(w_R^*(\eta); \eta)$. Denote this maximum value by

$$\Delta := R(0; \eta_R).$$

Then for all $w \geq 0$ and $0 < \eta < 1$, $R(w; \eta) \leq R(0; \eta_R) = \Delta$. This means if $\Delta \leq r/\lambda$, $R(w; \eta) \leq \Delta$, which means $D'(w) < 0$. (In fact, for any parameters, Δ is a constant number is approximately 0.1916.)

Part (ii). Fix $r/\lambda < \Delta$. The shape of $D(w)$ depends on η . There are three cases.

Case 1. By [Claim OA.2](#), for all $\eta \geq \eta_R$, $R(w; \eta)$ is decreasing in w for all $w \geq 0$ and is thus maximized at $w = 0$. Because $R(0; \eta)$ is decreasing in η , there exists $\bar{\eta} > \eta_R$ such that $R(0; \bar{\eta}) = r/\lambda$ and $R(0; \eta) < r/\lambda$ (and therefore $R(w; \eta) < r/\lambda$) for all $\eta > \bar{\eta}$. This implies $D'(w) \leq 0$ for all $w \geq 0$ and is thus maximized at $w = 0$ for all $\eta \geq \bar{\eta}$.

Case 2. By [Claim OA.2](#), for all $\eta < \eta_R$, $R(w; \eta)$ is single-peaked in w and is maximized at $w_R^*(\eta) > 0$. Because $R(0; \eta)$ is decreasing in η , there exists $\tilde{\eta} < \eta_R$ such that there exists a unique $w_{DM}^*(\eta)$ where $R(w_{DM}^*(\eta); \eta) = r/\lambda$, $R(w; \eta) < r/\lambda$ for $w < w_{DM}^*(\eta)$, and $R(w; \eta) > r/\lambda$ for $w > w_{DM}^*(\eta)$. This implies $D(w)$ single-peaked in w is uniquely maximized at $w = w_{DM}^*(\eta) > 0$ for $\eta \in [\tilde{\eta}, \bar{\eta})$.

Case 3. For all $\eta < \tilde{\eta}$, $R(0; \eta) < r/\lambda$ and $R(w; \eta)$ intersects with the line r/λ twice. Denote these two intersections by $w_L(\eta)$ and $w_H(\eta)$ with $w_L(\eta) < w_H(\eta)$. That is, $R(w_L(\eta); \eta) = R(w_H(\eta); \eta) = r/\lambda$.

Moreover, $R(w; \eta) < r/\lambda$ for $w < w_L(\eta)$ and $w \geq w_H(\eta)$, and $R(w; \eta) > r/\lambda$ for $w \in [w_L(\eta), w_H(\eta))$. This means that $D(w)$ decreases for $w \in [0, w_L(\eta))$, increases for $w \in [w_L(\eta), w_H(\eta))$, and decreases for $w \geq w_H(\eta)$. Therefore, for each $\eta < \tilde{\eta}$, $D(w)$ has two local maxima, one at $w = 0$ and one at $w = w_H(\eta)$. To determine which is the global maximizer, one can obtain $D(w_H(\eta))$ by plugging in the closed form expression for $w_H(\eta)$, and compare it with $D(0) = 1 - (1 - \eta)\eta$. It can be shown that there exists $\underline{\eta} < \tilde{\eta}$ such that $D(w_H(\eta)) < 1 - (1 - \eta)\eta$ for all $\eta < \underline{\eta}$, and $D(w_H(\eta)) \geq 1 - (1 - \eta)\eta$ for all $\eta \in [\underline{\eta}, \tilde{\eta})$. Then the optimal waiting time $w_{DM}^*(\eta) = 0$ for $\eta < \underline{\eta}$ and $w_{DM}^*(\eta) = w_H(\eta)$ for $\eta \geq \underline{\eta}$.

This concludes the proof. As an illustration, [Figure OA.1](#) plots the function $D(w)$ for several values of η .

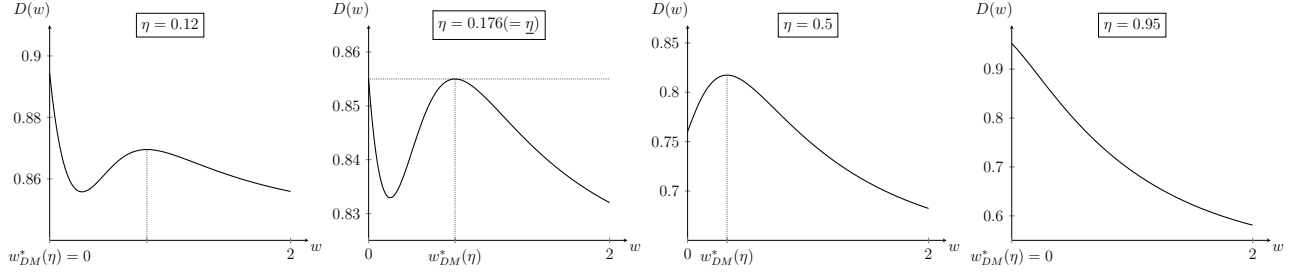


Figure OA.1: Decision maker's expected payoff from stopping at w $D(w)$ for $r = 0.5$, $\lambda = 5$, and $\beta = 0.5$. From left to right, the values of η are respectively $\eta = 0.12, 0.176, 0.5, 0.95$.